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Dear Scott:

I have reviewed the materials you forwarded to me recently regarding the Global Capitalism Index and Strength of Property Rights and Private Ownership Subindex. This note summarizes my general comments about the project, the two indexes, and some technical suggestions. The white paper is very good; the PowerPoint deck is helpful but raises questions (discussed below); and the subindex overview opens yet more questions. I know that this project is a work in progress and encourage you to simply put the best and most complete presentation forward. Overall, I conclude that this ambitious index project has the potential to advance significantly the understanding of scholars and policymakers about the definition, prevalence, determinants, diverse kinds, and trends of nation-level capitalism. I commend and support this project.

General comments

1. Audience. It will be a challenge to present a subject as rich as this index to a diverse audience of readers, including expert researchers (scholars), practitioners (such as policy makers, business analysts, or enterprise leaders), and consumers of research (such as students and the general public). The white paper does a good job addressing researchers. Practitioners and consumers will take the research details for granted and look for the “so what?” in the results. As you prepare for a public launch of the indexes, consider preparing some messaging appropriate for each segment of the audience. The white paper serves as a technical appendix for researchers that addresses the fine points raised in the methodology. Practitioners and consumers would benefit from a discussion that addresses how to use and interpret the results. More about this in the following sections.
2. Motivation. The white paper does a nice job of arguing that capitalism is poorly understood and that existing indexes are inadequate, hence the need for this new and better index. When you have produced the final version of the indexes, it would be interesting to provide a side-by-side comparison of the results of this index with those of Heritage and Fraser. The white paper could deepen the motivation for the indexes in two additional ways:
 - a. Application to teaching. In the early days, you and I envisioned an app that would make it easy for students to conduct a data-driven exploration of capitalism—discovery through data analysis. We test-taught some exercises to this effect. In an [article](#) for the *Journal of Economic Education*, I discussed the usefulness of data analysis in building knowledge, skills, and wisdom. The white paper and other materials neglect the instructional potential of the indexes. I suggest inserting some discussion along these lines.
 - b. Link to outcomes. So much of the argumentation for and against capitalism has to do with its association with outcomes about income and wealth, health, education, unemployment, equity, systemic stability, innovation, and governance (democracy vs.

autocracy). Arguably, this improved index will strengthen research and the resulting claims about criticism and advocacy. The index will also make it easier for students to develop their own insights by exploring the association between capitalism and outcomes.

3. Principal components analysis. It makes sense to use this technique for identifying the factors that explain the largest variation in the data. PCA is useful not only for identifying central tendencies, but also for detecting outliers (something that historians care about). Yet I would advise caution in how you present the methodology. First, the derived components can be difficult to interpret (or even name) because each is an amalgam of original variables. Second, since the methodology assumes linear relationships among variables, I worry about (1) nonlinearities such as threshold effects due to critical tipping points, (2) jump functions due to external shocks or structural shifts, (3) diminishing returns to scale, and (4) heteroskedasticity (a bigger problem in financial data) all of which linear models will not fit successfully. And finally, given the number of components feeding into the subindex and global index, you might face the problem of overfitting that could limit the ability of the model to generalize well. The white paper gives a good technical discussion of PCA, though I wish it had more to say about items 1-4 and gave more argument about why the linearity of PCA gives a reasonable starting point for analysis.
4. Promotion→Adoption→Impact. How will you measure the impact of the index project? Impact will depend on adoption of the indexes by researchers, practitioners, and consumers. Rough measures of adoption would be number of downloads by kind of audience, number of research papers using the indexes, website “hits,” number of cross-listings to other websites, etc. And adoptions follow from an effort to promote the indexes as research tools. How will you promote the indexes?
5. Definition of capitalism. The white paper and PowerPoint deck define “capitalism” in interesting and helpful ways.
 - a. “A system of institutions that privileges the ability to deploy capital and engage in market activity as freely as possible, absent constraints from both governmental and private actors.” (Page 2 of the PowerPoint deck, and page 3 of the white paper)
 - b. “At the core of our definition of capitalism is the understanding that well-functioning institutions are essential components of capitalism, not simply corrective adjustments to market adjustments to market externalities or dysfunction. ...Capitalism is not merely the absence of government intervention or the outright removal of constraints on private actors, but a system that depends on institutions to define, enforce, and stabilize economic interaction.” (White paper, page 5)

I see the importance of the institutional perspective (and agree with it) to distinguish this index from Heritage and Fraser, who take a more libertarian view. But I wish your definition had more grounding in “market activity” and open market prices as the defining features of capitalism. Branko Milanovic argued in a book that China is a capitalist country. Surely, China and North Korea can claim to have plenty of institutions for deploying capital and managing market activity, but does that make them capitalist? The U.S., with its layers of subsidies, tax breaks, regulations, and mandates is at best a “mixed economy”—would the U.S. qualify as capitalist under your definition? In an earlier blog post (see

<https://blogs.darden.virginia.edu/brunerblog/2024/10/what-is-capitalism-why-definitions-matter/>)

I argued for a definition based on voluntary exchange and open-market prices as the key attributes of capitalism.

- c. “**Capitalism.** An economic system based on voluntary exchange among buyers and sellers in which prices and other information drive the allocation of resources, competition and specialization promote efficiency, self-interest motivates decision-making, and personal investment creates private property, all of which are sustained by civil rights and institutions.”

I realize that a detailed discussion of definitions might be tiresome. But because the whole index project tends to argue for the existence of *many* kinds of capitalism, some good benchmark definition of capitalism would seem necessary.

A few detailed comments

- A. Concentration of data sources. V-Dem data series account for six of eight series for “Lack of Corruption,” and five of seven for “Rule of Law.” I would feel more comfortable if the data sources for these two components were more broadly diversified. In other sub-indexes, I also note concentrations (e.g. in the Market-Supporting Policy index, of the 22 data series, 19 are drawn from WEF).
- B. The seven components. The selection of the seven topical components for property rights and private ownership seem appropriate, though they also seem likely to overlap. For instance, rule of law, lack of corruption, and contract enforcement would seem likely to yield similar assessments for the various countries. Can you sharpen the rationale for distinguishing these three from each other? Presumably the PCA methodology corrects for multicollinearity among these components. Also, I note that the five most important variables all cluster within just three of the components (see PowerPoint pages 6 and 7)—does this imply that the other four components do not add meaningfully to the subindex?
- C. Market-Supporting Policy index. It is hard for me to tell from the titles of the data series, but the titles seem to imply that government intervention in the private sector is beneficial. In emerging economies especially, such intervention is a mixed blessing (see work by Hernando de Soto and William Easterley). I would be interested to see measures of the *size* of intervention relative to some basis such as GDP, industrial production, etc. Can you not find measures of state intervention by extent of business regulation, of state ownership in the private sector, of antitrust enforcement, and of incentives by the state in the form of cash subsidies (e.g., agricultural support), favorable (below-market) credit terms, monopolies granted, or tax breaks? Some possibly useful measures might be:
 - a. World Bank, time required to obtain an operating license. <https://data.worldbank.org/indicator/IC.FRM.DURS>
 - b. World Bank, subsidies and other transfers. <https://data.worldbank.org/indicator/GC.XPN.TRFT.ZS>
 - c. OECD MAGIC. <https://www.oecd.org/en/topics/subsidies-and-government-support.html> also see <https://cepr.org/voxeu/columns/government-subsidies-large-manufacturing-firms-insights-oecd-magic-database>
- D. State stability index. You might consider sourcing estimates of stability from more sources—especially private analysts, who have some economic “skin in the game”. See, for instance:
 - a. World population review. <https://worldpopulationreview.com/country-rankings/political-risk-index-by-country>
 - b. Willis political risk index. <https://www.wtwco.com/en-us/insights/2025/12/political-risk-index-h2-2025>
 - c. PRS Group. <https://www.prsgroup.com/regional-political-risk-index/>

Of special relevance to measures of state stability would be estimates of the political risk premium on the return required by investors across countries, and the default risk premium on bonds issued by national governments. Campbell Harvey (Duke) and Geert Bekaert (Columbia) have written several papers relevant to country risk. Some measures would include these:

 - d. Kroll country risk measures <https://www.kroll.com/en/tools-and-platforms/cost-of-capital/international-cost-of-capital>
 - e. Aswath Damodaran Data set at NYU https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/ctryprem.html
 - f. Metricgate. <https://metricgate.com/docs/country-risk/>
- E. Labor Market Openness index. It would be useful to reflect measures of the *exercise* of labor rights. Under labor protections, could you add a measure of days lost to strikes or other labor actions?
- F. Capital Market Sophistication index. When I hear experts discuss the leadership of the U.S. in international finance, they almost always cite the breadth and depth of the U.S. capital markets. It is not clear from the measures listed under this subindex that they capture this notion. For

instance, can you include a measure of the percent of global capital market trading activity claimed by each country?

- G. Ordinal vs. Interval vs. Categorical data. I understand the requirement to convert the many different data series to a common (interval) scale. But a close reading of some of the data series invites doubts about such conversion. Page 17 of the PowerPoint cites the inclusion of a banking crisis dummy, and page 11 of the subindex memo states that the "Expert survey question [is] posed as "Does the state own or directly control important sectors of the economy?" These two measures involve a categorical response rather than an interval (scaled) response. How does the transformation of data into a common scale deal with categorical data? Is this a pervasive issue in the various data series?
- H. Interpretation and significance. The broader issue in the previous point deals with the interpretation of the index estimates: page 26 of the PowerPoint deck gives the comparison of average index results (2008-2023) for the U.S. (global index 74.64, rank 10) with nine other countries having index scores in the 70s (ranks 5 to 15). But these index values seem so close together as to render the differences in ranking possibly meaningless. Indeed, the Global Capitalism Index datasheet shows the confidence interval upper and lower bounds of the US and New Zealand overlap, as do other country pairs in the zone of ranks 5-15. The uncertainty around index values makes it less meaningful to declare that a country has a particular rank and instead invites an acknowledgement that a country falls into a certain group, such as a decile of all countries. What matters is the *statistical significance* of the difference in index values among countries. Presumably, the confidence interval upper and lower bounds for each country would afford a way to test for the significance of index differences among countries or groups of countries. It would be interesting to see a table of pairwise significance test statistics for all pairs of countries in the sample.
- I. Missing data. The documents briefly mention an adjustment for missing data but does not explain the adjustment. Looking at the property rights data sheet, missing data is an endemic problem. Researchers will want to know more detail about treatment of missing data.
- J. Regions and other aggregates of countries. Given the existence of zones of economic mutual interest (such as the European Union, COMESA, NAFTA, APAC, and Mercosur) and geographic neighborhoods (such as Scandinavia, the Middle East, and the former states of the USSR in central Asia), it would interest your various audiences to see index values for such aggregates.
- K. Editing. Page 41, paragraphs 1 and 2 of the white paper appear to duplicate a sentence. Also, in some places, the white paper cites 178 countries in the global sample, and in others it cites 175 countries.

Conclusion

Let me stop here: you asked for 1-2 pages, and I've given 4. As Mark Twain supposedly said, "I didn't have time to write you a short letter, so I wrote you a long one." I hope these brief comments and suggestions are helpful and sufficiently clear and have tried to accommodate the breadth of review you requested, regarding theory, rigor, methodology, and execution. After addressing the comments above, I believe that this index project can make a substantive contribution to our understanding of capitalism and its stages of development in countries around the world and encourage you and your team toward its conclusion.

Sincerely,



Robert F. Bruner